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Juxtaposed and superimposed paleomagnetic primary and secondary components from the folded middle carboniferous sediments in the Reggane Basin (Saharan craton, Algeria)

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Abstract

A paleomagnetic study was carried out on the Middle Carboniferous sediments of the eastern margin of the Reggane Basin of Algeria. Seven sites (108 samples) in the Lower Serpukhovian and 11 sites (129 samples) in the Upper Serpukhovian, Bashkirian and Lower Moscovian levels were investigated.

Besides a common, but generally limited, viscous remanent magnetization (component A) and a recent chemical remanent magnetization of reversed polarity (A′), two main components were identified: one of these (component B), is characterized by a negative fold test and has been identified as a Lower Jurassic remagnetization. The associated paleomagnetic pole obtained in the seven zones by combining characteristic remanent magnetization directions (ChRM) and great circles ($\lambda = 71.1°$N, $\varphi = 251.4°$E, $A_{95} = 3.8°$, $K = 254$) lies in the vicinity of the NW African poles of similar ages. The second (component C) displays both normal and reversed polarities. Also determined by the combination of ChRM or stable end points and remagnetization circles, it yields a positive fold test which constrains the magnetization acquisition time and a positive reversal test which argues in favor of a "non-composite" nature of the component C. The normal polarities observed in the Lower Serpukhovian levels represent the latest normal event observed in Africa before the Kiaman superchron. The paleomagnetic South pole calculated from 10 sites ($n = 64$ data) gathered in four large areas ($\lambda = 26.5°$S, $\varphi = 44.7°$E, $A_{95} = 4.7°$, $K = 383$) is the first African Carboniferous pole founded on both positive reversal and fold tests. It lies only slightly apart from other Middle Carboniferous poles previously published for the northern part of Africa where no intraformational test were available to constrain the magnetization age. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetic tests; primary magnetization; paleomagnetic poles; Africa; Carboniferous

1. Introduction

Two previous paleomagnetic studies were carried out in the central Sahara of Algeria on red beds from

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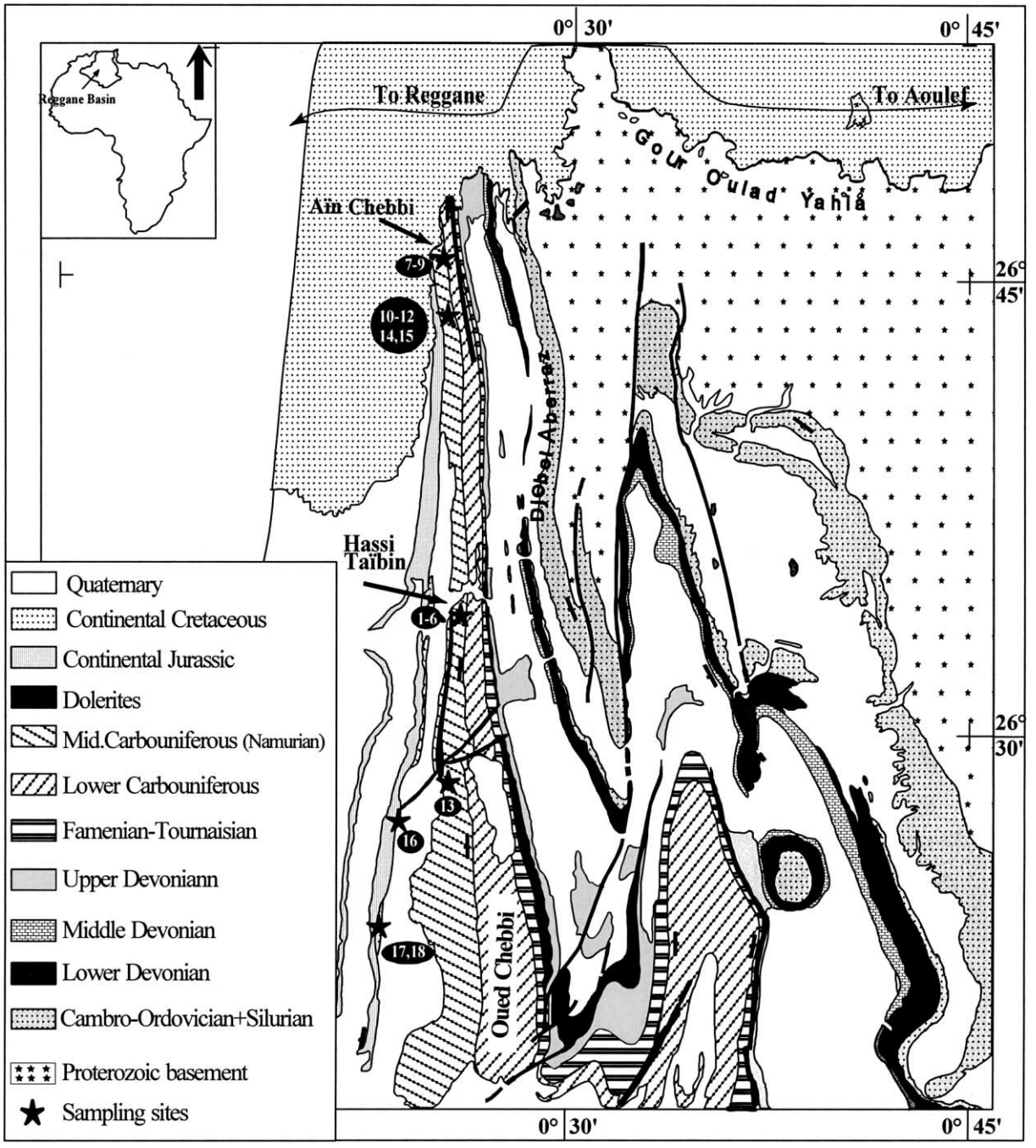


Fig. 1. Location of the sampling sites (stars) in the Middle Carboniferous formations, modified from the geological map of Reggane (A. Bensalah, S. Beuf, O. Gariel, G. Philippe, R. Lacot, A. Paris, D. Basseto, J. Conrad and A. Moussine-Pouchkine, DMG and Sonatrach, 1971–1972. Carte géologique de l'Algérie, 1/200,000).

Table 1
Geographic coordinates of the sites and their tilt parameters. Az.ddd: azimuth of the down dipping direction

Site	Lat. (°N)	Long. (°W)	Dip	Az.ddd
1	26°34′	0°26′	28	279
2	26°34′	0°26′	26	269
3	26°34′	0°26′	30	272
4	26°34′	0°26′	30	276
5	26°34′	0°26′	30	269
6	26°34′	0°26′	37	272
7	26°46′	0°25′	63	264
8	26°46′	0°25′	72	274
9	26°46′	0°25′	72	274
10	26°45′	0°25′	53	265
11	26°45′	0°25′	54	267
12	26°45′	0°25′	53	268
13	26°28′	0°25′	21	271
14	26°27′	0°24′	63	262
15	26°27′	0°24′	57	263
16	26°27′	0°24′	30	280
17	26°25′	0°24′	25	270
18	26°25′	0°24′	20	280

the upper part of the Middle Carboniferous, each on a limited number of samples. In the area of Aïn Chebbi and Hassi Taïbin, Conrad and Westphal (1973) analyzed two samples of red argillite from the Aïn Chebbi Formation (Conrad, 1984) of Lower Moscovian age. In the same region, south of Hassi Taïbin, Daly and Irving (1983) obtained significant results on 12 samples from four sites in the same formation. However, they could not perform the fold test because the four sites monotonously dip toward the west. In the red bed Formation of Hassi Bachir of about equivalent but slightly less constrained age (Conrad, 1984), Daly and Irving (1983) studied 31 samples from seven sites. Again no fold test was carried out because the easterly dip changes only very little from one site to the other. In the eastern border of the Reggane Basin, the Serpukhovian series (Djebel Berga limestones Formation and Hassi Taïbin gypsum Formation — Conrad, 1973, 1984) are marine with facies looking relatively similar to the Moscovian El Adeb Larache Formation of the Illizi Basin, from which a reliable paleomagnetic pole was obtained (Henry et al., 1992). Therefore we undertook a paleomagnetic study on these Serpukhovian levels and extended it to the red formation of Aïn Chebbi up to the Lower Moscovian levels. In order to determine a more reliable Middle-Carboniferous pole for Africa, the sampling included a greater number of sites covering a larger area with various dips allowing a fold test to be performed.

2. Geological setting

The studied area is located in the central Sahara of Algeria, at the eastern border of the Reggane Basin. The Precambrian basement is unconformably overlain by Paleozoic and Mesozoic deposits (Fig. 1). In the area, the Phanerozoic cover is folded in an N–S trending asymmetric anticline and the series outcrop along N–S bands. The dips are steep in the north in the Aïn Chebbi area (~70°) and gradually decrease toward the south where they do not exceed ~20° around Oued Chebbi (Table 1). The studied Middle Carboniferous formations are in continuity with the Visean levels but unconformably covered by an azoic continental formation attributed to the Upper Jurassic (Conrad, 1972, 1984). These continental Jurassic rubefied fluviatile conglomerates are themselves sealed disconformably by tabular Lower Cretaceous deposits.

The Middle Carboniferous series is subdivided into two formations. The lower part (limestones of the Djebel Berga Formation, surmounted by the Hassi Taïbin gypsum Formation — Conrad, 1973), is constituted by a succession of fossiliferous marine clays and fine limestones of Serpukhovian age, according to Conrad and Legrand-Blain (1971) and Conrad (1984). The upper part (Aïn Chebbi Formation) is mainly constituted by marine conglomeratic deposits at the bottom, followed by red sandstones and clays. The facies are dominantly continental, with few intercalated marine carbonate levels, which contain microfauna of Serpukhovian age (biozone H) at the bottom and of Early Moscovian age (Vereyian) at the top (Conrad et al., 1980; Legrand-Blain, 1983).

The Paleozoic series are crosscut by doleritic dykes and sills of Lower Jurassic age (Conrad, 1972). Two main tectonic phases affected this region and folded the Middle Carboniferous series (Conrad, 1981). The first one is post-Moscovian and even post-Stephanian according to Conrad (1984), because in the central part of the Reggane Basin the red continental formation, reached by drilling, extends conformably to the Wesphalian D and very likely up to the Stephanian

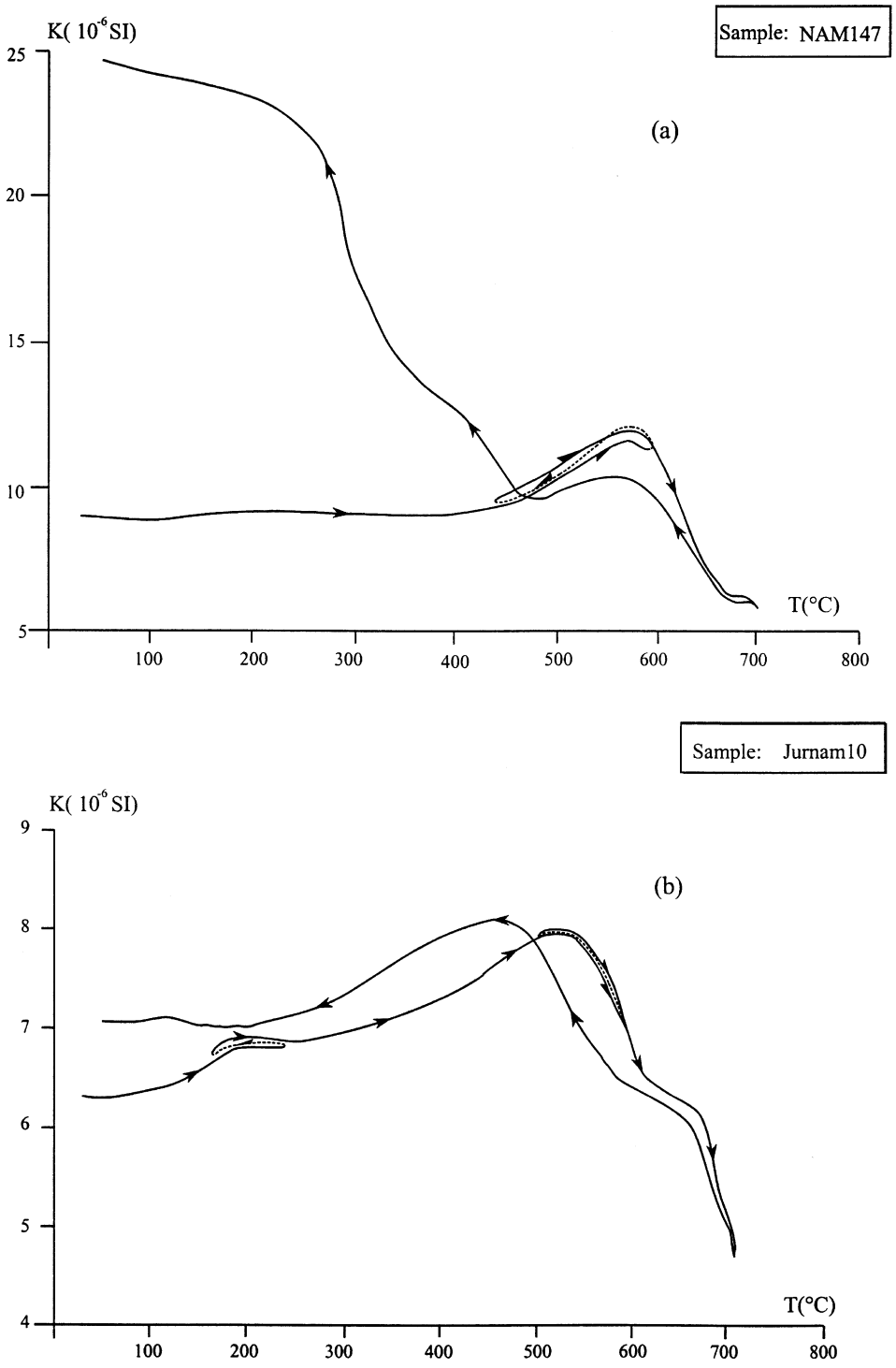


Fig. 2. Typical thermomagnetic curves (performed in the air) pointing out the presence of: (a) hematite; and (b) magnetite and hematite.

according to Bonnet et al. (1960). This first phase could thus be Permian, coeval with the Hercynian orogeny, which affected the Béchar Basin, the Ougarta mountains (Donzeau, 1972; Fabre, 1976) and the Maghrebine chain further north. The second one occurred after the continental presumably Upper Jurassic layers and prior to the deposition of the tabular continental Lower Cretaceous.

3. Sampling and analysis procedure

Two hundred and sixty five oriented cores distributed over 18 sites (Fig. 1, Table 1) have been collected in the Middle Carboniferous outcropping levels: sites 1–7 in the Formations of both the Djebel Berga limestones and the Hassi Taïbin gypsum, and sites 8–18 in the Aïn Chebbi Formation.

One to three specimens of standard size (cylinders of 11 cm^3) were cut from each sample, generally allowing both demagnetization treatments and additional rock magnetic studies to be performed. Prior to any demagnetization analysis, the specimens were stored in zero magnetic field for at least 1 month in order to reduce a possible viscous magnetization. The remanent magnetization was measured using a CTF cryogenic magnetometer or a JR4 spinner magnetometer (Agico, Brno). Preliminary tests on pilot samples from all the sampled area allowed to discard a few sites where the magnetization vectors were extremely low and noisy (several Lower Serpukhovian levels) or the directions unstable and inconsistent during the demagnetization process. Finally, 237 specimens were thermally demagnetized generally up to 675°C. In order to isolate and identify the magnetization components correctly, numerous steps were performed with increments ranging from 100°C in the lowest temperatures to 10°C in the highest ones. The magnetic susceptibility (K) was measured at room temperature after each heating step in order to monitor possible mineralogical changes having occurred upon heating.

The demagnetization process is presented on orthogonal vector plots (Wilson and Everitt, 1963; Zijderveld, 1967). The vectors remaining after each heating step and the difference vectors removed between two consecutive demagnetization steps are plotted on equal area projections. When clearly identified, the direction of the magnetization component was calculated by principal component analysis (Kirschvink, 1980), otherwise remagnetization circles (Halls, 1976, 1978; McFadden and McElhinny, 1988) were used when two adjoining magnetization components had partly or totally overlapping unblocking temperatures spectra. Mean characteristic directions were calculated using Fisher's statistics (Fisher, 1953) and a bivariate form of Fisher's statistics (Le Goff, 1990; Le Goff et al., 1992). Statistic parameters were also calculated during progressive unfolding of the structures in particular the precision parameter k (Fisher, 1953) and, for the elliptical confidence zone in the bivariate statistics, the value of $k\delta$ in the direction perpendicular to the fold axis (Henry and Le Goff, 1994).

4. Rock magnetism

In order to investigate the magnetic mineralogy of the studied samples, thermo-magnetic cycles were run in air, in several representative samples, using CS2 and Kappabridge KLY2 (Agico). In all these curves the presence of hematite was obvious (Fig. 2a). However, the red beds of Site 14 yield the most significant results; indeed besides hematite, magnetite appears as an important magnetic mineral (Fig. 2b). This magnetite is probably a pre-existing mineral, because no significant mineralogical alteration occurs upon heating as suggested by the relatively good reversibility of the thermomagnetic curves. Unfortunately, in other sites, strong mineralogical transformations were observed generally at around 350°C. They mostly correspond to the formation of magnetite, as suggested by the Curie temperature of the cooling curves. In these samples, the same trend is generally observed in the room temperature susceptibility: K remains stable up to 350–400°C where a weak decrease can be observed. After heating from 500 to 580°C, the susceptibility strongly increases, most likely due to the formation of magnetite. For samples giving significant magnetization directions in the high-temperature ranges, these directions do not change significantly upon demagnetization up to 675°C, so that the magnetic carriers of the remanence are believed to be unaffected by these mineralogical alterations.

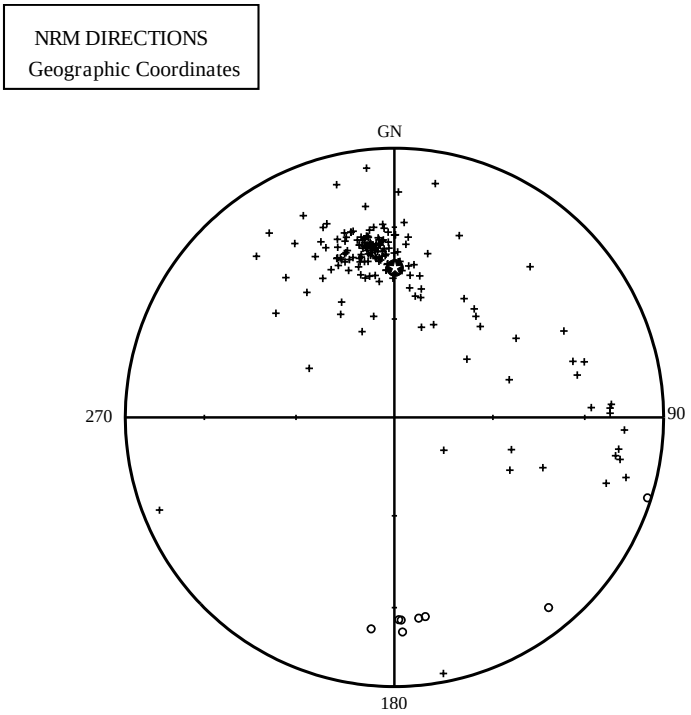


Fig. 3. Equal area plot of the NRM directions in geographic coordinates (crosses: positive inclinations, open circles: negative inclinations). White star: present geocentric dipole field direction.

5. Paleomagnetic analysis

As expected for the two families of rocks, the natural remanent magnetization (NRM) intensities are not homogeneous. Relatively high in the Aïn Chebbi red continental beds (around 22.5×10^{-7} Am^2 kg^{-1}), the intensity is about an order of magnitude lower in the marine limestones of the Djebel Berga Formation

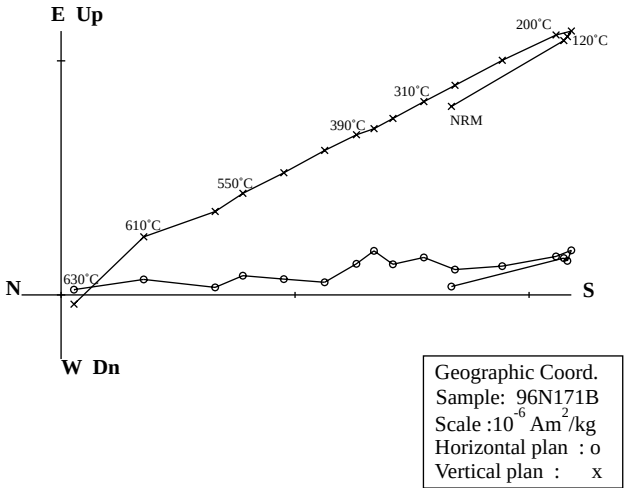


Fig. 4. Orthogonal vector plot for a sample displaying the component A′ (open circles: horizontal plane, crosses: vertical plane).

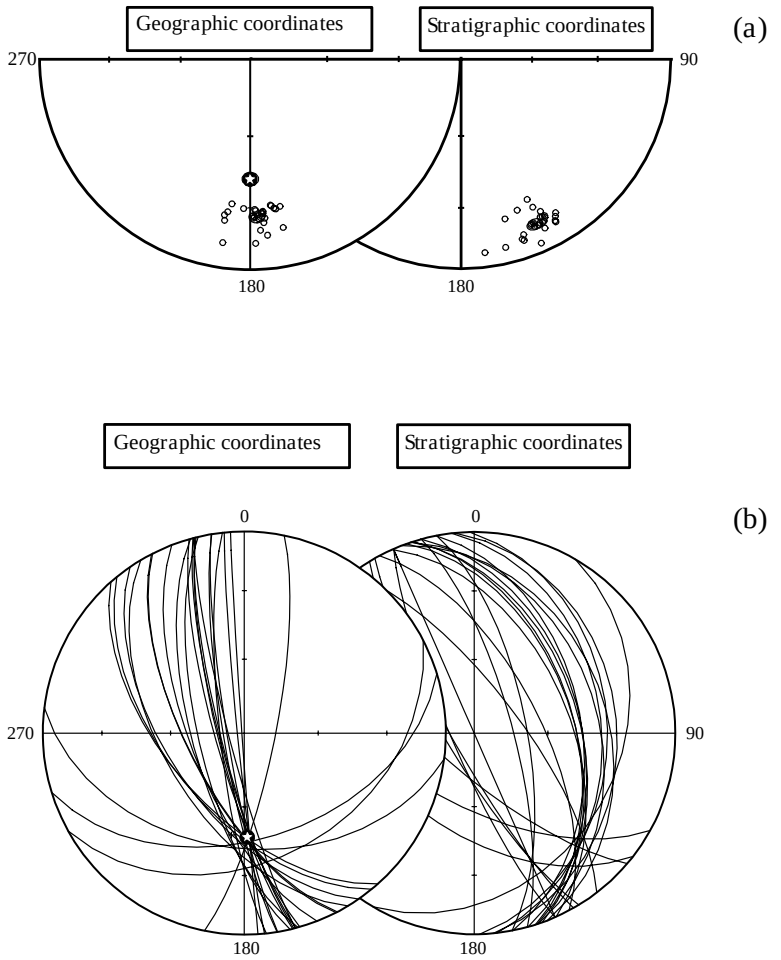


Fig. 5. Equal area plot of component A′: (a) ChRMs; and (b) (in situ great circles running toward the present earth field direction with reverse polarity — Upper hemisphere).

(around $2.8 \times 10^{-7}\,\mathrm{Am^2\,kg^{-1}}$) with large variations from a level to another.

Most of the NRM vectors are concentrated around and a little west of the present earth field (PEF) direction (Fig. 3). We note however that several directions are scattered within a girdle elongated between NW and ESE directions. A few directions also lie near the present Earth's magnetic field direction with reversed polarity. This arrangement clearly indicates the existence of more than one component in the NRM, which needs to be unraveled in order to isolate the primary remanent magnetization.

During thermal demagnetization, after elimination of a viscous component A (sometimes important in the lowest temperature ranges), two behaviors of the magnetic vectors were observed: either a linear component, or in few cases a stable end point could be determined, or a remagnetization circle could be fitted to the data in case of overlapping unblocking temperatures spectra of two adjacent magnetization components. Depending on the sites and sometimes on the samples, two main magnetization components could be isolated (B and C) together with several directions apparently well defined by linear segments on the orthogonal plots (component A′) or intermediate between two end members. The question arises as to whether these directions represent real component or artifacts.

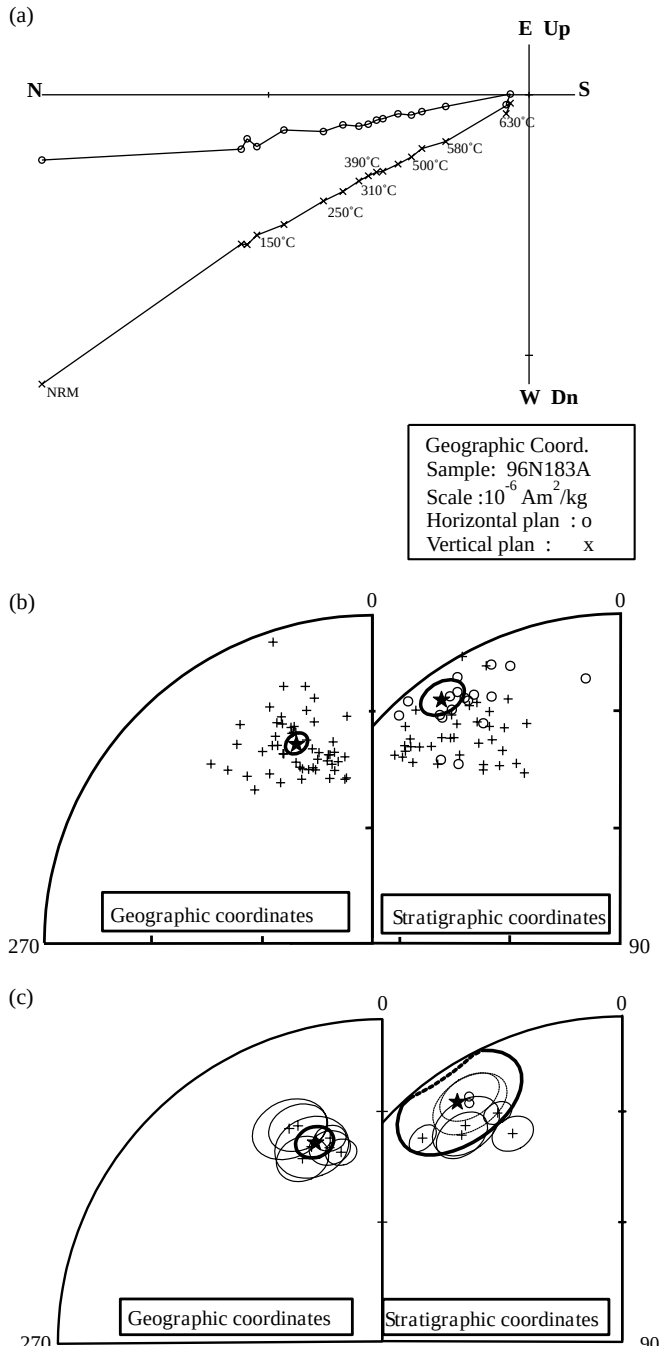


Fig. 6. (a) Orthogonal vector plot for a sample displaying the B component (open circles: horizontal plane, crosses: vertical plane); (b) equal area plot of the ChRMs directions of component B, in geographic and stratigraphic coordinates, star: mean direction with its 95% confidence cone; and (c) equal area plot of the mean component B direction calculated in the seven zones by combining ChRMs and great circles directions according to McFadden and McElhinny (1988), star: mean direction and associated 95% confidence cone.

Table 2
Average directions of component B calculated at each zone by combining ChRMs and great circles. N' is the total number of data combined, n and N are, respectively, the number of ChRMs and great circles used, Declination D and Inclination I are in degrees, α_{95} and k are Fisher's parameters, λ and φ are the latitude and the longitude of the calculated paleomagnetic pole, respectively, and A_{95} and K their corresponding Fisher's parameters

	Individual sites regrouped	N' ($n+N$)	Before dip correction				After dip correction				λ (°N)	φ (°E)	dp,dm A_{95} (°)	K
			D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)				
ZONE 1	7+14+15	9 (7+2)	347.2	39.4	225	3.5	315.4	13.3	263	3.2	77.6	252.4	2.5; 4.2	
ZONE 2	8	11 (11+0)	338.5	28.9	53	6.4	327.9	−12.1	53	6.4	67.0	245.2	3.9; 7.1	
ZONE 3	9	8 (8+0)	336.3	28.7	48	8.1	327.2	−13.9	48	8.1	65.2	247.7	4.9; 8.9	
ZONE 4	1+2	11 (4+7)	345.2	35.0	185	3.5	331.3	21.0	244	3.1	74.6	245.4	2.3; 4.0	
ZONE 5	3+4	12 (10+2)	336.3	37.8	56	5.9	322.0	20.6	55	5.9	67.7	261.6	4.1; 7.0	
ZONE 6	5+6	6 (6+0)	339.8	35.7	67	8.2	323.9	19.0	79	7.6	70.3	254.5	5.5; 9.5	
ZONE 7	13+16+18	13 (8+5)	344.4	37.5	85	4.6	332.1	28.5	81	4.7	74.7	252.8	3.2; 5.4	
Mean 7 zones		7	341.0	34.8	208	4.2	325.6	11.1	21	13.5	71.0	251.4	2.8; 4.8	
Mean 7 poles		7									71.1	251.4	3.8	254

5.1. Component A′

A component A′ with southerly declination and slightly less inclined than the reversed present-day dipole field direction (Fig. 4) is observed in 28 samples from six sites (Fig. 5a). These directions are well defined in the intermediate and high unblocking temperature ranges (200–300°C ≤ T_{ub} ≤ 580–620°C). The average direction of this component is, for $n = 28$ characteristic remanent magnetization directions (ChRMs): $D = 177.6°$, $I = -25.7°$, $\alpha_{95} = 2.5°$, $k = 122$ and $D = 156.0°$, $I = -15.1°$, $\alpha_{95} = 2.5°$, $k = 122$ before and after dip correction, respectively. As the majority of the samples come from the same area with similar dips, the tectonic correction does not provide any information here. We note that this A′ component is always associated to a PEF component, generally a viscous remanent magnetization (VRM) but also sometimes of chemical remanent magnetization (CRM) origin, of normal polarity and spanning the lowest unblocking temperatures. From samples without stable direction, if we trace great circles running toward the A′ direction (Fig. 5b), their intersection in geographic coordinates: $D = 177.2°$, $I = -41.3°$ for $N = 23$ circles lies only 4.1° aside from the reverse dipole field direction ($D = 180°$, $I = -44.9°$). After tectonic correction the circles are more scattered and the intersection ($D = 147.1°$, $I = -18.8°$) is poorly defined which demonstrates that this component is a remagnetization.

Two interpretations can be proposed for this component A′: it may correspond to a Lower Eocene remagnetization, because the paleomagnetic pole calculated from the 28 ChRM from four sites without dip correction ($\lambda = 76.4°$N, $\varphi = 185.7°$E, $A_{95} = 3.8°$, $K = 593$) falls on the 50 My average pole of the Apparent Polar Wander Path (APWP) of Besse and Courtillot (1991) for Africa. But as it does not correspond to any known tectonic or pedogenetic event in this area, we rather favor the idea that we are probably dealing with a reversed CRM of recent origin, roughly opposite to the present dipole field direction, as indicated by the intersection of the great circles. However, as the ChRM directions never perfectly reach this intersection direction we suggest that the A′ component could be a composite one in which some of the unblocking temperatures of PEF directions with normal polarity overlap with those of a slightly older component of reversed polarity.

5.2. Component B

Component B is the most common one (Fig. 6a); the directions lie a little west from the present dipole field direction in geographic coordinates (Fig. 6b). It is found in almost all the sites (14) either as well-defined directions (ChRMs) or represented by great circles in the case of overlapping of two adjacent magnetization components when no ChRM could be isolated. It may or may not be associated with a limited VRM. In spite of a great variability from site to site as a result of rock type variations, the unblocking temperatures of this component most often span the intermediate and/or high-temperature ranges (200–300°C to 550–650°C). The mean ChRM direction isolated in 54 samples (Fig. 6b) from 11 sites is: $D = 339.3°$ $I = 35.4°$, $\alpha_{95} = 2.6°$, $k = 56$ and $D = 324.3°$, $I = 9.3°$, $\alpha_{95} = 4.9°$, $k = 17$, respectively, in geographic and in stratigraphic coordinates. We note that the dispersion increases upon untilting of the structures.

In order to characterize this component better at the site level we combined, wherever possible, the ChRMs directions and the informations brought by the great-circle analysis (McFadden and McElhinny, 1988). Each sample was used only once, either as ChRM or as circle. When the number of data per site (ChRMs or circles) was too limited to allow a VGP to be calculated, the samples were grouped by sites of the same zone with almost identical dips. We end up with 70 data from seven zones (Fig. 6c and Table 2) in which the number of samples is relatively homogeneous, ranging from 6 to 13. The mean B component direction calculated in this way becomes for $N = 7$ zones: $D = 341.0°$, $I = 34.8°$, $\alpha_{95} = 4.2°$, $k = 208$ and $D = 325.6°$, $I = 11.1°$, $\alpha_{95} = 13.5°$, $k = 21$, respectively, before and after tectonic correction (Table 2). It is not significantly different from the mean direction obtained on the ChRMs alone, and clearly points out the failure of the fold test and the secondary nature of this component. A progressive unfolding of the structures according to Le Goff (1990) and Le Goff et al. (1992) confirms that the best grouping of the directions is reached for no tectonic correction at all and that component B is post-tectonic. The mean pole (Table 2) calculated from the seven paleomagnetic poles in geographic coordinates ($\lambda = 71.1°$N, $\varphi = 251.4°E$, $A_{95} = 3.8°$,

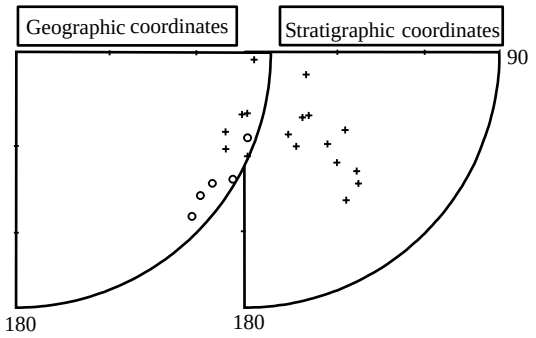


Fig. 7. Equal area plot of the directions intermediate between the PEF or B directions and the C component, before and after dip correction.

$K = 254$) is close to the Early (200 and 180 My) Jurassic pole of the synthetic APWP curve (for Africa) of Beck (1994) and is also in agreement with the APWP of Besse and Courtillot (1991). It lies 5.6° east from the average pole for Early Jurassic calculated by Kosterov and Perrin (1996) for NW Africa. As the Paleozoic series is intruded by doleritic sills of Lower Jurassic age (Fig. 1) contemporaneous with the early stage of the Atlantic opening (Conrad, 1972) and because these intrusions are themselves tilted, this result suggests that the B remagnetization component is related to the post-dolerite tectonic phase reported by Conrad (1981) which likely occurred after the magmatic event.

5.3. "Intermediate" directions

Anticipating the existence of a final primary C component which will be discussed below, a few words must be said about a group of 11 directions determined in Site 14 for unblocking temperatures lower than 580°C. They are spread in the south eastern quadrant with inclinations close to zero in geographic coordinates and steep to intermediate after dip correction (Fig. 7).

It is worth noting that this population displays an elongated distribution in intermediate position between the PEF or B components and the C directions in stratigraphic coordinates. They correspond to unblocking temperatures of magnetite (between 400 and 580°C, Fig. 8a–c). In the examples (given in Fig. 8a–c) the direction carried by magnetite seems to be well determined and slightly distinct from the C component carried by hematite. However, we generally note in the orthogonal plots of other samples, and on the same T_{ubs} range, a slight curvature in the intervals selected to determine this component. This is not always obvious because the directions are often scattered; in such a case, the directions lie along great circles joining either the PEF or the B directions and the reversed C component determined in the highest temperature ranges (600–670°C). Thus, the elongated distribution of these directions between known components suggests to interpret these "intermediate" directions as composite ones resulting from an imperfect separation of the previously defined components (PEF or B, and C) and consequently we will not consider them any further. Similar composite components determined by principal component analysis have also been observed recently in two formations of the Tindouf Basin (Henry et al., 1999; Merabet et al., 1999).

5.4. Component C

Component C is constituted by two groups of apparently antipodal polarity (Fig. 9a). The main group has SE declinations with positive inclinations while the other displays directions with NW declinations and negative inclinations.

The directions are revealed in a total of 64 samples in 10 sites either as ChRM or as remagnetization circles. Most of them have been determined in temperature ranges where no significant change of the susceptibility measured at room temperature occurs, i.e. prior to mineralogical alterations. For the other samples, the remanent directions do not change notably even when a significant increase of the susceptibility is observed.

5.4.1. ChRMs

In stratigraphic coordinates (Fig. 9a), most of the samples ($n = 26$) distributed in six sites (Table 3) display SE declinations with positive inclinations (SE group of reverse polarity). Nine other samples from two sites have NW declinations with negative inclinations (NW group of normal polarity).

In the SE group, the C component is generally well characterized by linear segments on orthogonal plots

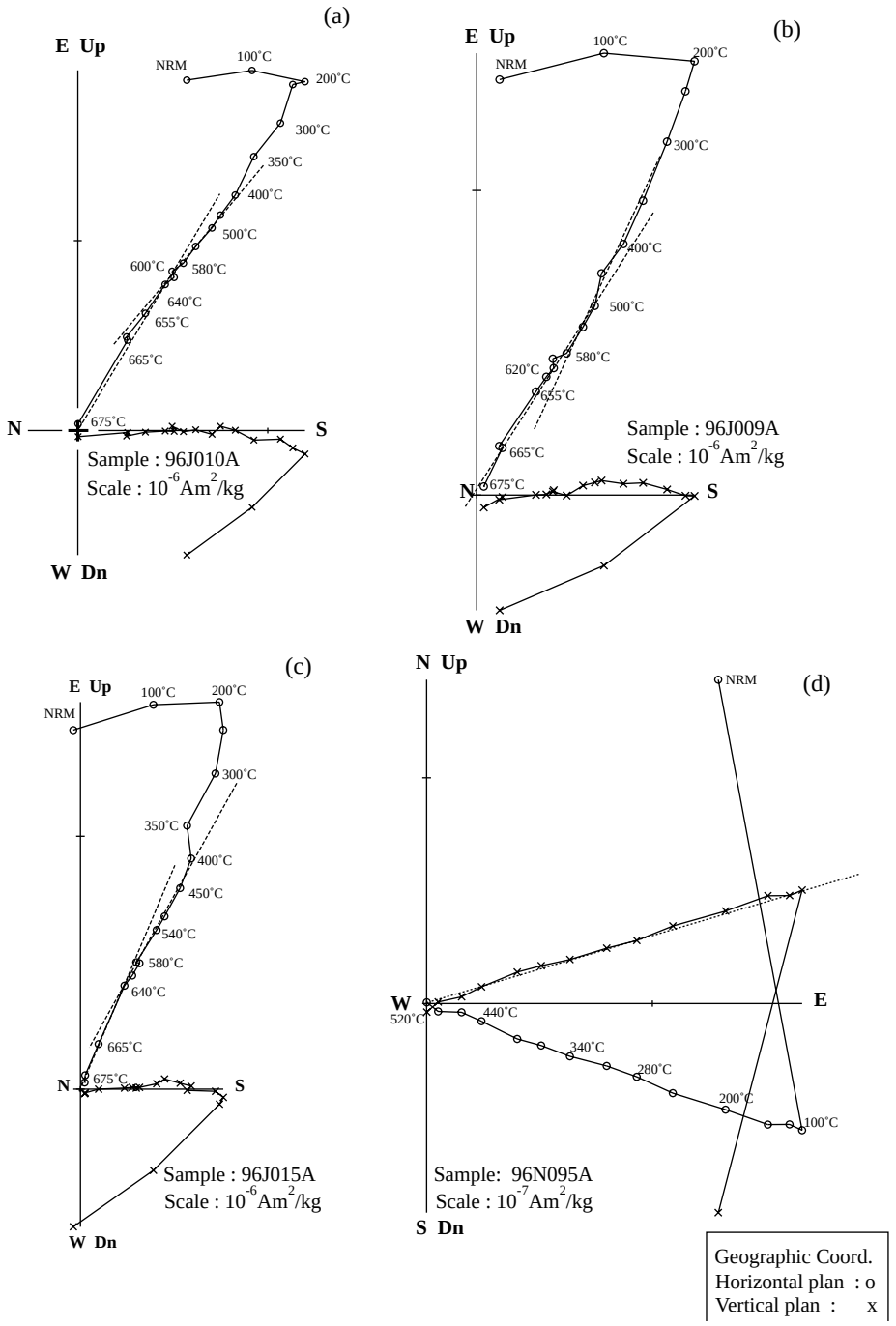


Fig. 8. Orthogonal vector plots for samples displaying: (a)–(c) the composite direction and the C component (carried by heamatite) with reverse polarity; and (d) the C reverse component carried by magnetite, (open circles: horizontal plane, crosses: vertical plane).

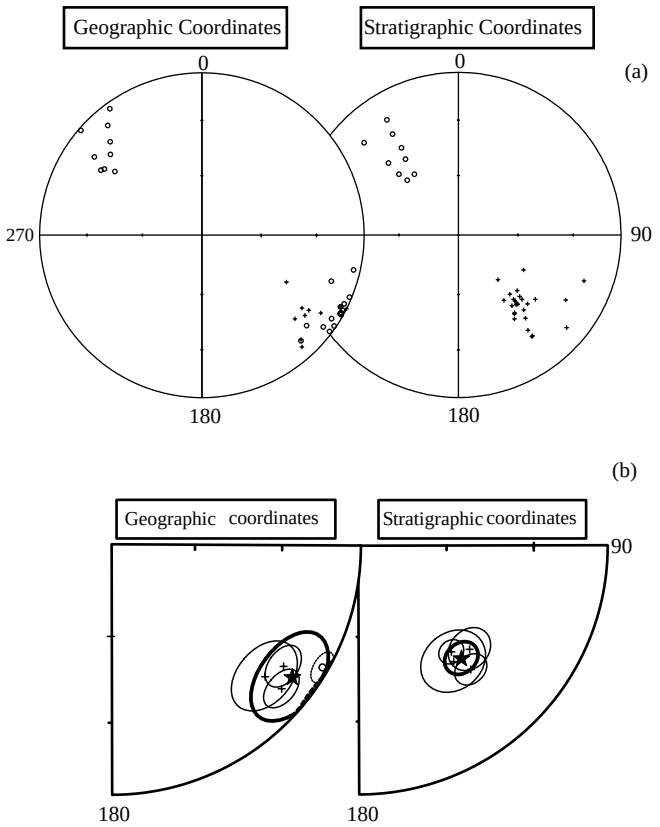


Fig. 9. Equal area plot of: (a) the ChRM directions of the C component, before and after tectonic correction; and (b) the mean component C direction calculated in the four areas before and after tectonic correction, respectively (all converted to a reversed polarity). The star represents the mean direction and its associated 95% confidence cone.

Table 3
Mean ChRM directions of component C obtained in each individual site (*n*: number of samples) before and after dip correction, and corresponding precision parameters, *k*, and α_{95}

			Before Dip Correction				After Dip Correction			
Site	Polarity	*n*	*D* (°)	*I* (°)	*k*	α_{95} (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)
Lower serpukhovian										
1	Normal	1	304.4	−27.8			316.9	−51.9		
	Reversed	1	123.2	13.4			129.6	38.4		
2	Normal	8	312.4	−17.6	37	9.2	322.2	−35.1	37	9.2
5	Reversed	1	118.0	−2.0			121.0	24.0		
6	Reversed	2	120.2	−16.3			120.3	16.5		
Upper Serpukhovian–Lower Moscovian										
14	Reversed	15	120.0	−2.7	94	4.0	139.0	42.9	94	4.0
16	Reversed	2	127.0	22.6			138.9	48.0		
17	Reversed	5	130.7	21.1	32	13.7	141.2	38.7	32	13.7

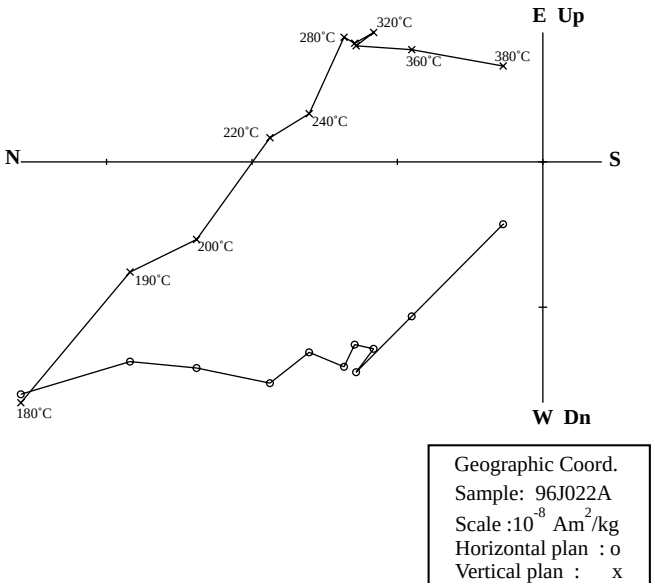


Fig. 10. Orthogonal vector plot for a sample displaying (at high temperature) the C component with normal polarity. For more clarity, (NRM intensity strongly decreased after heating at 100°C) the NRM and the thermal 100°C steps have been removed from this diagram (see text), (open circles: horizontal plane, crosses: vertical plane).

(Fig. 8a–c) and is carried by hematite in the red bed levels as attested by the high unblocking temperatures interval in which the ChRMs are determined. In the other (Serpukhovian) samples this component is less well defined, generally by stable end points, between 300 and 520°C (Fig. 8d), suggesting that the magnetic carrier is magnetite.

The mean direction of the SE group is for $n = 26$ ChRMs: $D = 122.5°$; $I = 3.4°$; $\alpha_{95} = 5.6°$; $k = 27$ in geographic coordinates, and $D = 136.5°$; $I = 39.8°$; $\alpha_{95} = 4.6°$; $k = 38$ in stratigraphic coordinates.

In the NW group, the C component (Fig. 10) is found only in Lower Serpukhovian levels. The magnetization intensity decreases always abruptly at 100°C suggesting the presence of goethite carrying a strong VRM or CRM and about only 1/10 of the magnetization intensity remains after 180°C treatment. The C component could only be determined by stable end points between 280 and 450°C (in best cases).

The average direction in the NW group is for $n = 9$ ChRMs: $D = 311.6°$; $I = -18.8°$; $\alpha_{95} = 8.4°$; $k = 38$ in geographic coordinates and $D = 321.8°$; $I = -37.0°$; $\alpha_{95} = 8.8°$; $k = 35$ in stratigraphic coordinates.

5.4.2. Combination with the great circles

In spite of the large collection of samples, most of them do not allow to define a ChRM for component C. However a total of 25 samples, on which the final ChRM could not be reached, yield great circles evolving from the B or the PEF direction toward the C component: in most cases (20 samples), this evolution is toward the SE group (Fig. 11a) while in five samples it is toward the NW group (Fig. 11b). In the sites where ChRM C were obtained for other samples, the great circles always go through these directions, which again justifies the combination of ChRMs and great circles according to McFadden and McElhinny (1988).

For the reverse group, using all the ChRMs ($n = 26$ samples) and the great circles ($N = 20$) on other samples, the paleomagnetic direction is: $D = 127.0°$, $I = 6.0°$, $\alpha_{95} = 4.3°$, $k = 25$ and $D = 136.1°$, $I = 39.5°$, $\alpha_{95} = 3.5°$, $k = 38$ before and after tectonic correction, respectively.

For the normal group, combining $n = 9$ ChRM and $N = 5$ great circles, the average direction is: $D = 312.6°$, $I = -19.1°$, $\alpha_{95} = 6.3°$, $k = 42$ and $D = 323.0°$, $I = -37.0°$, $\alpha_{95} = 6.5°$, $k = 39$, before and after dip correction, respectively.

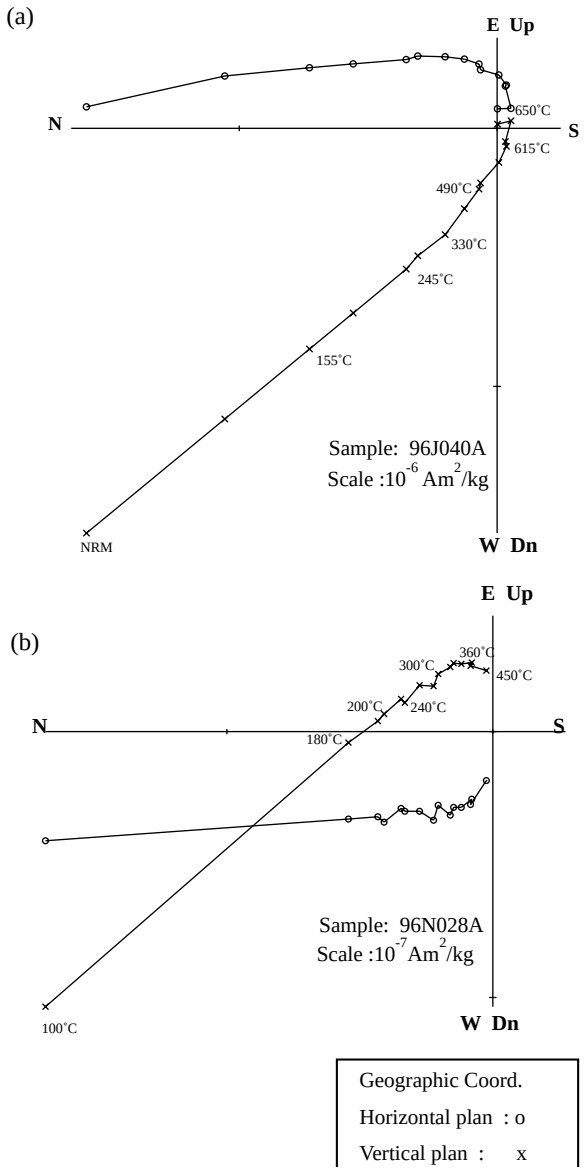


Fig. 11. Orthogonal vector plots for samples displaying: (a) superimposed components with direction running toward the SE; and (b) superimposed components with direction running toward the NW (open circles: horizontal plane, crosses: vertical plane).

5.4.3. Reversal test

On these two populations constituted by a total of 14 data for the normal polarity and 46 for the reverse one, we applied the reversal test of McFadden and McElhinny (1990). Before dip correction, the critical angle γ_c is 7.4° and the observed angle between the two mean polarities is $\gamma_o = 14.2°$. The test is here clearly negative.

After dip correction, the critical angle γ_c is 6.3° and the observed angle between the two mean polarities becomes $\gamma_o = 6.0°$. The reversal test is thus positive and classified as class B according to McFadden and McElhinny (1990). This result indicates that each one of two directions with opposite polarity indeed corresponds to a single component and argues for a positive fold test.

5.4.4. Fold test

Although the studied formations in the eastern border of the Reggane Basin outcrop in a west dipping monoclinal structure, a fold test can be applied here because the dips are highly variable from north to south. The fold test of Le Goff (1990), Le Goff et al. (1992) and Henry and Le Goff (1994) allows us to calculate the variation of various statistic parameters during progressive unfolding.

This test was performed on the 35 ChRM converted to a unique polarity. The variation of $k\delta$ (Henry and Le Goff, 1994) which is, in the bivariate form of Fisher's (1953) statistics, the value of the precision parameter in the direction perpendicular to the fold axis, shows that the best grouping of the directions is obtained for 87% unfolding. As the maximum of the $k\delta$ values is spread around 87%, there is no statistical significant difference between 87 and 100% unfolding. We note that if we apply the simple and stringent McElhinny's (1964) method using the $k\delta$ values instead of the classical k Fisher parameter, the test is positive for 100% unfolding. This confirms that the fold test on the C component is positive.

5.4.5. Paleomagnetic directions

In this study, only a limited number of samples per site yields either significant C ChRMs (Tables 3 and 4) or great circles (Table 4). In order to calculate a reliable pole corresponding to the C component, we gathered together the different neighboring sites of roughly similar dips to constitute four areas. The first area is close to Hassi Taibin (six sites: from 1 to 6, Fig. 1) and corresponds to the sites of Lower Serpukhovian age; the second one is located around Aïn Chebbi (eight sites). The third area (sites 13 and 16) is located northwest of Oued Chebbi and the

Table 4
Mean directions of the C component in each area, calculated by combination of ChRMs and great circles before and after tectonic correction. Four data from Daly and Irving (1983) have been included as ChRMs (three in Area 3 and one in Area 4). n and N are, respectively, the number of ChRM and great circles (GC) used, λ and φ are the south pole coordinates, K and A_{95} the corresponding Fisher's parameters

Areas	n (ChRM)	N (GC)	Before dip correction				After dip correction				λ (°S)	φ (°E)	A_{95} (°)	K
			D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	A_{95} (°)				
1	13	14	130.1	11.9	25	5.7	137.9	33.4	32	5.1	29.4	47.4		
2	15	0	120.0	−2.7	94	4.0	139.0	42.9	94	4.0	24.9	41.4		
3	5	8	125.1	16.9	49	6.1	133.1	39.0	39	6.9	23.6	48.0		
4	6	3	130.5	20.2	25	10.7	140.8	39.7	27	10.3	28.1	41.8		
Mean direction(four areas)	39	25	126.4	11.6	52	12.8	137.7	38.8	297	5.3	26.6	44.7		
Mean pole (four areas)											26.5	44.7	4.7	383

fourth one (sites 17 and 18) is situated further south, west of Oued Chebbi.

Areas 2, 3 and 4 correspond to Upper Serpukhovian to Lower Moscovian ages. In order to complete our data, we included the results obtained by Daly and Irving (1983) in the Aïn Chebbi region. Their component, determined from stable end points after heating at 600°C, corresponds to our ChRM C. Because they only give average directions for each site, each one of their data was integrated in our mean directions per area (Table 4) as equivalent to one sample. Four ChRMs were thus added to our collection, three in Area 3 and one in Area 4.

The directions calculated after dip correction for the 13 Lower Serpukhovian samples (Table 3) of Area 1 ($D = 135.3°$, $I = 33.3°$, $k = 24$, $\alpha_{95} = 8.6°$) and for the 22 Upper Serpukhovian to Lower Moscovian samples (Table 3) of Areas 2, 3 and 4 ($D = 139.5°$, $I = 42.4°$, $k = 71$, $\alpha_{95} = 3.7°$) are not significantly different at the 95% confidence level. Therefore these data have been combined to compute a Middle Carboniferous paleomagnetic direction. Adding the great circles data, this direction for the four areas is (Table 4): $D = 126.4°, I = 11.6°, k = 52, \alpha_{95} = 12.8°$ and $D = 137.7°$, $I = 38.8°$, $k = 297$, $\alpha_{95} = 5.3°$, before and after dip correction, respectively.

We applied the fold test to the four areas (Fig. 9b). The average paleomagnetic directions of the four

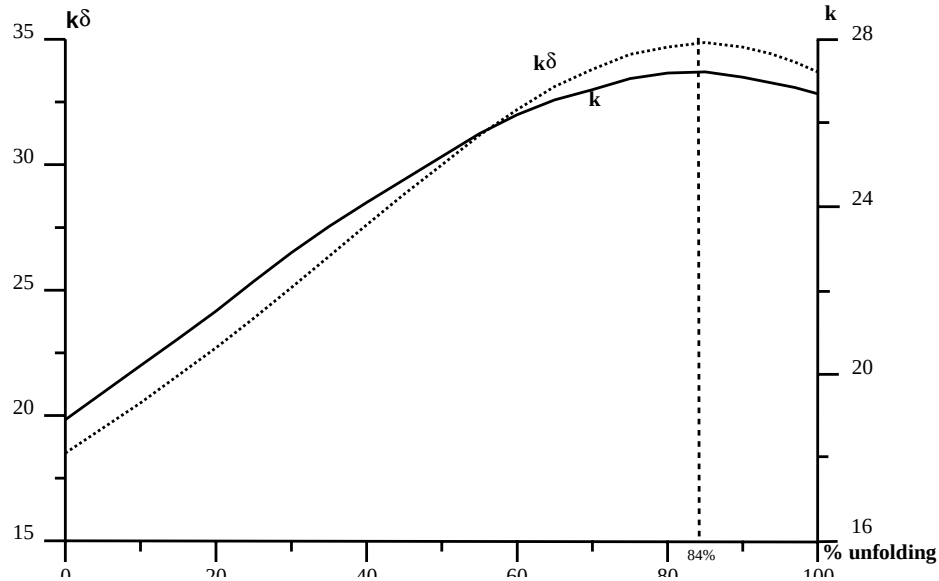


Fig. 12. Variation of $k\delta$ (Henry and Le Goff, 1994) and k (Fisher, 1953) parameters during progressive unfolding of the four areas.

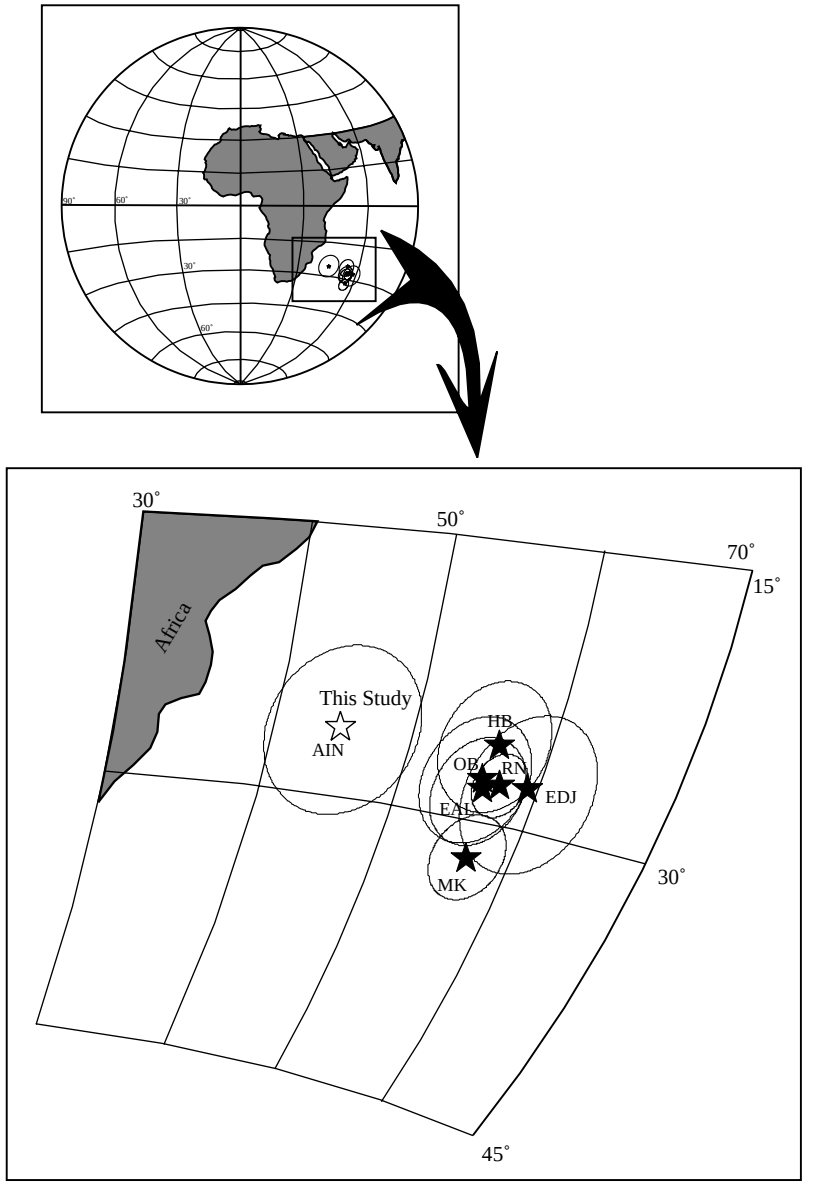


Fig. 13. North African Middle to Upper Carboniferous paleomagnetic poles (see Table 5).

areas (Table 4) yields a positive fold test when the McElhinny's (1964) method is used. The best values of the statistic parameters ($k\delta$, k Fisher and α_{95}) correspond to an unfolding of 84% (Fig. 12), but because of the small variation of these parameters between 84 and 100% unfolding, the fold test can be considered as positive.

The C component is thus characterized by both positive fold and reversal tests. The magnetization was therefore acquired before the first phase of folding which occurred after the Stephanian (Conrad, 1984). Therefore this component represents very likely a primary magnetization. The south paleomagnetic pole computed from the paleomagnetic poles calculated in each one of the four areas is: $\lambda = 26.5°$S, $\varphi = 44.7°$E, $A_{95} = 4.7°$, $K = 383$.

Table 5
Middle and upper carboniferous poles for stable Africa

	Age	Rock unit	Lat. (°S)	Long. (°E)	A_{95}	Ref.
RN	Namurian	Reouina	28.4	56.9	1.7	Merabet et al., 1999
OB	Bashkirian	Oubarakat	28.2	55.5	3.4	Derder et al., 2000a
HB	Upper Carboniferous	Hassi Bachir	26.8	56.6	3.7	Daly and Irving, 1983
EAL	Moscovian	El Adeb Larache	28.7	55.9	2.9	Henry et al., 1992
EDJ	Moscovian	Edjeleh	28.3	58.9	4.2	Derder et al., 2000b
MK	Lower Stephanian	Merkala	32.4	56.6	2.3	Henry et al., 1999
AIN	Middle Carboniferous	Aïn Ech Chebbi	26.5	44.7	4.7	This study

6. Discussion and conclusions

In the previous Middle-Carboniferous paleomagnetic studies from several regions in the Saharan craton (Daly and Irving, 1983; Henry et al., 1992; Merabet et al., 1999; Henry et al., 1999; Derder et al., 2000a,b), only paleomagnetic directions of reversed polarity were found. In the present study, the Upper Serpukhovian to Lower Moscovian levels also yield reversed polarity, but the paleomagnetic directions display both normal and reversed directions in the Lower Serpukhovian. It is the first time that normal magnetization directions are observed in Middle-Carboniferous rocks of Africa. It constitutes thus the most recent normal polarity event observed in Africa before the Kiaman reversed superchron and confirms that the Kiaman period could start as early as the end of the Serpukhovian (inside the biozone H).

In none of these previous studies was the time of magnetization acquisition constrained by any paleomagnetic test, except for the Moscovian Formation at Edjeleh where a positive fold test was obtained (Derder et al., 2000b). In the present study, the single component nature of the characteristic direction was demonstrated by a reversal test, and its age constrained by the fold test. Our paleomagnetic pole (26.5°S, 44.7°E) places this part of Africa in the southern hemisphere in subtropical paleolatitudes (around 18–24°S), which is consistent with biological observations (reefs further north in the Béchar region — Lemosquet et al., 1976) and with lithological evidence (evaporites in the Reggane and Ahnet Basins). For all the above reasons we believe that this pole represents a reliable Middle Carboniferous data for Africa. It is in fairly good agreement with the other North African poles of similar ages (Fig. 13 and Table 5), but all the poles determined either east (Daly and Irving, 1983; Henry et al., 1992; Derder et al., 2000a,b) or west (Merabet et al., 1999; Henry et al., 1999) of the region of the present study are tightly grouped together (average pole at $\lambda = 28.8°$ S, $\varphi = 56.7°$E, $A_{95} = 1.8°$, $K = 1420$) whereas our pole lies a little but significantly west from the other ones at the 95% confidence level (angular distance = 10.9°). Parts of our sites have very low dip and this gap cannot be explained by local structures and maladapted structural correction at Ain Chebbi. The main difference between our data and the previous ones concerns inclination calculated for a same site from the different paleomagnetic poles. In fact, inclination appears steeper for our pole. Assumption of an "inclination shallowing" existing in all the other sites in a uniform manner seems less credible. A difference in the age of magnetization acquisition should better justify the difference of the paleomagnetic poles. In fact, in all the previous studies, only magnetizations of reversed polarity have been found (like during Kiaman period). In the studied area, the presence of both polarities could indicate a different age at least for the oldest beds. Moreover, looking at the general apparent drift of the Gondwanian poles (see for example Chen et al., 1994), our pole could be older than the other poles. This suggests that the magnetization was acquired in our sites before the other sites. That could be related to an effective difference in age of the formations (but some of the other sites are also in pre-Moscovian formations). An alternative hypothesis could be a widespread remagnetization having affected all the sites except our new one (Bayou et al., 2000). This remagnetization, however, should have occurred before the end of the Carboniferous because African Stephano-Autunian poles are different from the Moscovian ones of the previous studies, and a positive

fold test (for pre-Stephano-Autunian folding) has been obtained at Edjeleh (Derder et al., 2000b). Such remagnetization should have occurred in a window of time including part of the Moscovian and of Stephanian, i.e. relatively few after deposition of the formations. It was not evidenced in pre-Carboniferous formations of the Saharan craton, except surprisingly perhaps in the Reggane Basin close to the area studied here (Bayou et al., 2000). The present paleomagnetic data set does not allow a choice between these possible explanations.

Acknowledgements

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